TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	70.7 ± 2.19
Correlation Stability	90.4 ± 1.56
Distribution Consistency	83.6 ± 5.82
Entropy Preservation	66.6 ± 0.49
Mean Z Score Drift	0.1 ± 0.03
Categorical Fidelity	0.0 ± 0.00
Privacy Risk	100.0 ± 0.00
Data Density	100.0 ± 0.00
Coverage Score	0.0 ± 0.00
Feature Accuracy	93.9 ± 2.26
Novelty Score	0.0 ± 0.00

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Dataset Audit: parkinsons.csv

1. Dataset Portrait

Attribute	Value
Rows	195
Columns	23
Numeric features	23
Categorical features	0
Missing cells (%)	0.0%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	gpt-4o-mini
Folds (*13)	10
Context % (*14)	20%
Generation % (*15)	100%
Join context (*16)	True

3. Statistical Significance & Data Drift

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	70.7 ± 2.19
Correlation Stability (*2)	90.4 ± 1.56
Distribution Consistency (*3)	83.6 ± 5.82
Entropy Preservation (*4)	66.6 ± 0.49
Mean Z Score Drift (*5)	0.1 ± 0.03
Categorical Fidelity (*6)	0.0 ± 0.00
Privacy Risk (*7)	100.0 ± 0.00
Data Density (*8)	100.0 ± 0.00
Coverage Score (*9)	0.0 ± 0.00
Feature Accuracy (*10)	93.9 ± 2.26
Novelty Score (*11)	0.0 ± 0.00

*[1] (*12) Algorithm: Contextual constraint or model parameter.*

*[2] (*13) Folds: Contextual constraint or model parameter.*

*[3] (*14) Context %: Contextual constraint or model parameter.*

*[4] (*15) Generation %: Contextual constraint or model parameter.*

*[5] (*16) Join context: Contextual constraint or model parameter.*

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9333333333333333	N/A	0.9333333333333333	0	
F1	0.9560674056986365	N/A	0.9560674056986365	0	

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9153846153846154	N/A	0.9230769230769231	0.007692307692307675	
F1	0.9425432250595183	N/A	0.947891480490723	0.00534825543120473	

Model: DT

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8615384615384617	N/A	0.853846153846154	-0.007692307692307698	
F1	0.9030929149651324	N/A	0.8985587866792273	-0.004534128285905192	

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8102564102564103	N/A	0.8102564102564103	0	
F1	0.8848220246893213	N/A	0.8848220246893213	0	

Model: LR

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8692307692307694	N/A	0.8692307692307694	0	
F1	0.9130547168372463	N/A	0.9130547168372463	0	

Model: AB

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Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9128205128205129	N/A	0.9128205128205129	0	
F1	0.941802264596966	N/A	0.941802264596966	0	

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7153846153846153	N/A	0.7153846153846153	0	
F1	0.765535606787701	N/A	0.765535606787701	0	

Model: KNN

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.8358974358974359	N/A	0.8358974358974359	0	
F1	0.8946275603661134	N/A	0.8946275603661134	0	

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7692307692307693	N/A	0.8025641025641026	0.03333333333333332	
F1	0.8530601056701009	N/A	0.8759992829794054	0.02293917730930454	

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.9256410256410257	N/A	0.9205128205128206	-0.005128205128205121	
F1	0.95092541040947	N/A	0.947314395198361	-0.00361101521110882 57	

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	MDVP:Fo	MDVP:Fhi	MDVP:Flo	MDVP:Jitter	MDVP:Jitter.1	MDVP:RAP	MDVP:PPQ	Jitter:DDP	MDVP:Shimmer	MDVP:Shimmer.1
1	119.992	157.302	74.997	0.00784	0.00784	0.0037	0.00554	0.01109	0.04374	0.04374
2	122.4	148.65	113.819	0.00968	0.00968	0.00465	0.00696	0.01394	0.06134	0.06134
3	116.682	131.111	111.555	0.0105	0.0105	0.00544	0.00781	0.01633	0.05233	0.05233
4	116.676	137.871	111.366	0.00997	0.00997	0.00502	0.00698	0.01505	0.05492	0.05492
5	116.014	141.781	110.655	0.01284	0.01284	0.00655	0.00908	0.01966	0.06425	0.06425
6	120.552	131.162	113.787	0.00968	0.00968	0.00463	0.0075	0.01388	0.04701	0.04701
7	120.267	137.244	114.82	0.00333	0.00333	0.00155	0.00202	0.00466	0.01608	0.01608
8	107.332	113.84	104.315	0.0029	0.0029	0.00144	0.00182	0.00431	0.01567	0.01567
9	95.73	132.068	91.754	0.00551	0.00551	0.00293	0.00332	0.0088	0.02093	0.02093
10	95.056	120.103	91.226	0.00532	0.00532	0.00268	0.00332	0.00803	0.02838	0.02838
11	88.333	112.24	84.072	0.00505	0.00505	0.00254	0.0033	0.00763	0.02143	0.02143
12	91.904	115.871	86.292	0.0054	0.0054	0.00281	0.00336	0.00844	0.02752	0.02752
13	136.926	159.866	131.276	0.00293	0.00293	0.00118	0.00153	0.00355	0.01259	0.01259
14	139.173	179.139	76.556	0.0039	0.0039	0.00165	0.00208	0.00496	0.01642	0.01642
15	152.845	163.305	75.836	0.00294	0.00294	0.00121	0.00149	0.00364	0.01828	0.01828
16	142.167	217.455	83.159	0.00369	0.00369	0.00157	0.00203	0.00471	0.01503	0.01503
17	144.188	349.259	82.764	0.00544	0.00544	0.00211	0.00292	0.00632	0.02047	0.02047
18	168.778	232.181	75.603	0.00718	0.00718	0.00284	0.00387	0.00853	0.03327	0.03327
19	153.046	175.829	68.623	0.00742	0.00742	0.00364	0.00432	0.01092	0.05517	0.05517
20	156.405	189.398	142.822	0.00768	0.00768	0.00372	0.00399	0.01116	0.03995	0.03995

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B.1 Audit Trace: Fold 1 | Ctx: 20% | Gen: 100%

#	MDVP:Fo	MDVP:Fhi	MDVP:Flo	MDVP:Jitter	MDVP:Jitter.1	MDVP:RAP	MDVP:PPQ	Jitter:DDP	MDVP:Shimmer	MDVP:Shimmer.1
1	193.03	208.9	80.297	0.00766	0.00766	0.0045	0.00389	0.01351	0.03044	0.03044
2	206.327	226.355	92.02	0.00495	0.00495	0.00302	0.00246	0.00905	0.02105	0.02105
3	178.222	202.45	141.047	0.00321	0.00321	0.00163	0.00194	0.00488	0.03759	0.03759
4	154.003	160.267	128.621	0.00183	0.00183	0.00076	0.001	0.00229	0.0103	0.0103
5	209.144	237.494	109.379	0.00282	0.00282	0.00147	0.00152	0.00442	0.01861	0.01861
6	120.08	139.71	111.208	0.00405	0.00405	0.0018	0.0022	0.0054	0.01706	0.01706
7	110.417	131.067	103.37	0.00784	0.00784	0.00352	0.00514	0.01056	0.03715	0.03715
8	208.083	253.792	91.802	0.00757	0.00757	0.00428	0.00428	0.01285	0.06725	0.06725
9	244.99	272.21	239.17	0.00451	0.00451	0.00279	0.00237	0.00837	0.01897	0.01897
10	228.969	239.541	113.201	0.00238	0.00238	0.00136	0.0014	0.00408	0.01745	0.01745

B.2 Audit Trace: Fold 2 | Ctx: 20% | Gen: 100%

#	MDVP:Fo	MDVP:Fhi	MDVP:Flo	MDVP:Jitter	MDVP:Jitter.1	MDVP:RAP	MDVP:PPQ	Jitter:DDP	MDVP:Shimmer	MDVP:Shimmer.1
1	119.1	128.442	88.833	0.00692	0.00692	0.00393	0.003	0.01179	0.02682	0.02682
2	152.125	161.469	76.596	0.00382	0.00382	0.00191	0.00226	0.00574	0.05925	0.05925
3	148.79	158.359	138.99	0.00309	0.00309	0.00152	0.00186	0.00456	0.01574	0.01574
4	116.682	131.111	111.555	0.0105	0.0105	0.00544	0.00781	0.01633	0.05233	0.05233
5	128.94	479.697	88.251	0.00581	0.00581	0.00241	0.00314	0.00723	0.02008	0.02008
6	151.884	157.765	133.751	0.00258	0.00258	0.00115	0.00122	0.00346	0.0145	0.0145
7	245.51	262.09	231.848	0.00235	0.00235	0.00127	0.00148	0.0038	0.01608	0.01608
8	115.38	123.109	108.634	0.00332	0.00332	0.0016	0.00199	0.0048	0.01503	0.01503
9	95.056	120.103	91.226	0.00532	0.00532	0.00268	0.00332	0.00803	0.02838	0.02838
10	143.533	162.215	65.809	0.01101	0.01101	0.00647	0.00467	0.01941	0.05384	0.05384

B.3 Audit Trace: Fold 3 | Ctx: 20% | Gen: 100%

#	MDVP:Fo	MDVP:Fhi	MDVP:Flo	MDVP:Jitter	MDVP:Jitter.1	MDVP:RAP	MDVP:PPQ	Jitter:DDP	MDVP:Shimmer	MDVP:Shimmer.1
1	126.512	141.756	99.77	0.01936	0.01936	0.01159	0.0099	0.03476	0.06734	0.06734
2	136.969	166.607	66.004	0.00923	0.00923	0.00507	0.00463	0.0152	0.03111	0.03111
3	173.898	211.35	74.677	0.00448	0.00448	0.00237	0.00254	0.0071	0.06727	0.06727
4	217.116	233.481	93.978	0.00404	0.00404	0.00127	0.00128	0.00381	0.01299	0.01299
5	157.821	172.975	68.401	0.00358	0.00358	0.00196	0.00196	0.00587	0.03716	0.03716
6	108.807	134.656	102.874	0.00761	0.00761	0.00349	0.00486	0.01046	0.02719	0.02719
7	242.852	255.034	227.911	0.00225	0.00225	0.00117	0.00139	0.0035	0.01494	0.01494
8	139.224	586.567	66.157	0.03011	0.03011	0.01854	0.01628	0.05563	0.09419	0.09419
9	186.163	197.724	177.584	0.00298	0.00298	0.00165	0.00175	0.00496	0.01495	0.01495
10	116.879	131.897	108.153	0.00788	0.00788	0.00334	0.00493	0.01003	0.02645	0.02645

B.4 Audit Trace: Fold 4 | Ctx: 20% | Gen: 100%

#	MDVP:Fo	MDVP:Fhi	MDVP:Flo	MDVP:Jitter	MDVP:Jitter.1	MDVP:RAP	MDVP:PPQ	Jitter:DDP	MDVP:Shimmer	MDVP:Shimmer.1
1	125.641	141.068	116.346	0.03316	0.03316	0.02144	0.01522	0.06433	0.09178	0.09178
2	119.031	127.533	109.216	0.0044	0.0044	0.00214	0.00192	0.00641	0.01033	0.01033
3	145.174	198.109	80.637	0.00733	0.00733	0.00414	0.00422	0.01242	0.02362	0.02362
4	186.163	197.724	177.584	0.00298	0.00298	0.00165	0.00175	0.00496	0.01495	0.01495
5	117.274	129.916	110.402	0.00752	0.00752	0.00299	0.00469	0.00898	0.02293	0.02293
6	114.847	271.314	104.68	0.00867	0.00867	0.00373	0.0052	0.0112	0.03225	0.03225
7	208.519	220.315	199.02	0.00609	0.00609	0.00368	0.00339	0.01105	0.01761	0.01761
8	173.898	211.35	74.677	0.00448	0.00448	0.00237	0.00254	0.0071	0.06727	0.06727
9	116.014	141.781	110.655	0.01284	0.01284	0.00655	0.00908	0.01966	0.06425	0.06425
10	98.804	102.305	87.804	0.00432	0.00432	0.00247	0.00258	0.00742	0.02223	0.02223

B.5 Audit Trace: Fold 5 | Ctx: 20% | Gen: 100%

#	MDVP:Fo	MDVP:Fhi	MDVP:Flo	MDVP:Jitter	MDVP:Jitter.1	MDVP:RAP	MDVP:PPQ	Jitter:DDP	MDVP:Shimmer	MDVP:Shimmer.1
1	176.824	215.293	83.961	0.0046	0.0046	0.00209	0.00221	0.00628	0.01169	0.01169
2	116.682	131.111	111.555	0.0105	0.0105	0.00544	0.00781	0.01633	0.05233	0.05233
3	167.93	193.221	79.068	0.00442	0.00442	0.0022	0.00247	0.00661	0.04351	0.04351
4	180.978	200.125	155.495	0.00406	0.00406	0.0022	0.00244	0.00659	0.03852	0.03852
5	198.764	396.961	74.904	0.0074	0.0074	0.0037	0.0039	0.01109	0.02296	0.02296
6	117.004	144.466	99.923	0.00353	0.00353	0.00176	0.00218	0.00528	0.01657	0.01657
7	177.876	192.921	168.013	0.00411	0.00411	0.00233	0.00241	0.007	0.02126	0.02126
8	119.992	157.302	74.997	0.00784	0.00784	0.0037	0.00554	0.01109	0.04374	0.04374
9	116.556	592.03	86.228	0.00496	0.00496	0.00254	0.00263	0.00762	0.0166	0.0166
10	148.272	164.989	142.299	0.00459	0.00459	0.0025	0.00256	0.0075	0.0419	0.0419

B.6 Audit Trace: Fold 6 | Ctx: 20% | Gen: 100%

#	MDVP:Fo	MDVP:Fhi	MDVP:Flo	MDVP:Jitter	MDVP:Jitter.1	MDVP:RAP	MDVP:PPQ	Jitter:DDP	MDVP:Shimmer	MDVP:Shimmer.1
1	168.778	232.181	75.603	0.00718	0.00718	0.00284	0.00387	0.00853	0.03327	0.03327
2	203.184	211.526	196.16	0.00178	0.00178	0.00094	0.00106	0.00283	0.00958	0.00958
3	198.764	396.961	74.904	0.0074	0.0074	0.0037	0.0039	0.01109	0.02296	0.02296
4	260.105	264.919	237.303	0.00339	0.00339	0.00205	0.00186	0.00616	0.0203	0.0203
5	244.99	272.21	239.17	0.00451	0.00451	0.00279	0.00237	0.00837	0.01897	0.01897
6	121.345	139.644	98.25	0.00684	0.00684	0.00388	0.00332	0.01164	0.02534	0.02534
7	100.96	110.019	95.628	0.00606	0.00606	0.00351	0.00348	0.01053	0.02427	0.02427
8	240.301	245.135	219.783	0.00517	0.00517	0.00316	0.00283	0.00949	0.02018	0.02018
9	157.821	172.975	68.401	0.00358	0.00358	0.00196	0.00196	0.00587	0.03716	0.03716
10	152.125	161.469	76.596	0.00382	0.00382	0.00191	0.00226	0.00574	0.05925	0.05925

B.7 Audit Trace: Fold 7 | Ctx: 20% | Gen: 100%

#	MDVP:Fo	MDVP:Fhi	MDVP:Flo	MDVP:Jitter	MDVP:Jitter.1	MDVP:RAP	MDVP:PPQ	Jitter:DDP	MDVP:Shimmer	MDVP:Shimmer.1
1	167.93	193.221	79.068	0.00442	0.00442	0.0022	0.00247	0.00661	0.04351	0.04351
2	192.818	224.429	168.793	0.03107	0.03107	0.018	0.01958	0.05401	0.11908	0.11908
3	241.404	248.834	232.483	0.00281	0.00281	0.00157	0.00173	0.0047	0.0176	0.0176
4	146.845	208.701	81.737	0.00496	0.00496	0.0025	0.00275	0.00749	0.01919	0.01919
5	159.116	168.913	144.811	0.00342	0.00342	0.00178	0.00184	0.00535	0.03381	0.03381
6	148.462	161.078	141.998	0.00397	0.00397	0.00202	0.00235	0.00605	0.01831	0.01831
7	129.336	139.867	118.604	0.0049	0.0049	0.00165	0.00183	0.00495	0.02498	0.02498
8	120.256	125.306	104.773	0.00407	0.00407	0.00224	0.00205	0.00671	0.01516	0.01516
9	116.15	131.731	109.815	0.00381	0.00381	0.00181	0.00232	0.00542	0.03026	0.03026
10	117.87	127.349	95.654	0.00647	0.00647	0.00356	0.003	0.01067	0.03087	0.03087

B.8 Audit Trace: Fold 8 | Ctx: 20% | Gen: 100%

#	MDVP:Fo	MDVP:Fhi	MDVP:Flo	MDVP:Jitter	MDVP:Jitter.1	MDVP:RAP	MDVP:PPQ	Jitter:DDP	MDVP:Shimmer	MDVP:Shimmer.1
1	125.791	140.557	96.206	0.01378	0.01378	0.00826	0.00655	0.02478	0.04689	0.04689
2	153.046	175.829	68.623	0.00742	0.00742	0.00364	0.00432	0.01092	0.05517	0.05517
3	122.964	130.049	114.676	0.00428	0.00428	0.00124	0.00155	0.00373	0.01681	0.01681
4	150.44	163.441	144.736	0.00396	0.00396	0.00206	0.00233	0.00619	0.02551	0.02551
5	117.87	127.349	95.654	0.00647	0.00647	0.00356	0.003	0.01067	0.03087	0.03087
6	151.989	157.339	132.857	0.00174	0.00174	0.00075	0.00096	0.00225	0.01024	0.01024
7	100.96	110.019	95.628	0.00606	0.00606	0.00351	0.00348	0.01053	0.02427	0.02427
8	163.656	200.841	76.779	0.00742	0.00742	0.0038	0.0039	0.0114	0.01659	0.01659
9	148.09	162.824	67.343	0.00762	0.00762	0.00467	0.00354	0.014	0.05428	0.05428
10	153.848	165.738	65.782	0.0084	0.0084	0.00428	0.0045	0.01285	0.0381	0.0381

B.9 Audit Trace: Fold 9 | Ctx: 20% | Gen: 100%

#	MDVP:Fo	MDVP:Fhi	MDVP:Flo	MDVP:Jitter	MDVP:Jitter.1	MDVP:RAP	MDVP:PPQ	Jitter:DDP	MDVP:Shimmer	MDVP:Shimmer.1
1	114.238	124.393	77.022	0.00581	0.00581	0.00299	0.00316	0.00896	0.04009	0.04009
2	125.036	143.946	116.187	0.0128	0.0128	0.00743	0.00623	0.02228	0.03886	0.03886
3	121.345	139.644	98.25	0.00684	0.00684	0.00388	0.00332	0.01164	0.02534	0.02534
4	198.383	215.203	193.104	0.00212	0.00212	0.00113	0.00135	0.00339	0.01263	0.01263
5	214.289	260.277	77.973	0.00567	0.00567	0.00295	0.00317	0.00885	0.01884	0.01884
6	170.368	268.796	79.543	0.00571	0.00571	0.00232	0.00269	0.00696	0.03273	0.03273
7	126.144	154.284	97.543	0.00975	0.00975	0.00593	0.00454	0.01778	0.02852	0.02852
8	223.361	263.872	87.638	0.00352	0.00352	0.00169	0.00188	0.00506	0.02536	0.02536
9	112.014	588.518	107.024	0.00533	0.00533	0.00268	0.00329	0.00805	0.02448	0.02448
10	107.332	113.84	104.315	0.0029	0.0029	0.00144	0.00182	0.00431	0.01567	0.01567

TabuLLM: AI Benchmark Report

B.10 Audit Trace: Fold 10 | Ctx: 20% | Gen: 100%

#	MDVP:Fo	MDVP:Fhi	MDVP:Flo	MDVP:Jitter	MDVP:Jitter.1	MDVP:RAP	MDVP:PPQ	Jitter:DDP	MDVP:Shimmer	MDVP:Shimmer.1
1	126.144	154.284	97.543	0.00975	0.00975	0.00593	0.00454	0.01778	0.02852	0.02852
2	206.327	226.355	92.02	0.00495	0.00495	0.00302	0.00246	0.00905	0.02105	0.02105
3	113.166	130.27	100.673	0.00502	0.00502	0.00257	0.00312	0.00772	0.05279	0.05279
4	229.401	252.221	221.156	0.00205	0.00205	0.00114	0.00113	0.00342	0.01457	0.01457
5	150.258	154.609	75.349	0.00248	0.00248	0.00105	0.00136	0.00315	0.01131	0.01131
6	136.358	176.595	65.75	0.00971	0.00971	0.00534	0.00478	0.01601	0.04978	0.04978
7	128.001	138.052	122.08	0.00436	0.00436	0.00137	0.00166	0.00411	0.02297	0.02297
8	155.078	163.736	144.148	0.00168	0.00168	0.00068	0.00092	0.00204	0.01064	0.01064
9	157.821	172.975	68.401	0.00358	0.00358	0.00196	0.00196	0.00587	0.03716	0.03716
10	129.336	139.867	118.604	0.0049	0.0049	0.00165	0.00183	0.00495	0.02498	0.02498